Institute of Information Technology (IIT)

Noakhali Science and Technology University

Bachelor of Science in Software Engineering

Final Examination - 2022 (Year-4, Term-2)

Course Title: Software Maintenance **Total Time:** 3 Hours

Course Code: SE 4203 **Full Marks:** 70

Answer all the Seven (07) Questions from the following: 7 X 10 = 70

Scenario (Question 1-2)

ABC Tech Inc. has been contracted to develop a new inventory management system for Global Retail Chain (GRC). After ten months of development, the system was successfully delivered to GRC. For the pilot phase, three stores were selected to use the system. The feedback from these stores was overwhelmingly positive, with staff finding the system intuitive and efficient. Encouraged by the pilot results, GRC decided to implement the system across all its stores. However, after several months of use, numerous issues emerged. Employees reported that the system occasionally failed to process orders correctly, leading to stock discrepancies. Additionally, the system was found to be incompatible with some of the older hardware still in use at various store locations.

1\. (a) How does the scenario of ABC Tech Inc. developing an inventory management system for Global Retail Chain (GRC) illustrate the importance of understanding the context in software maintenance activities? 3

(b) Provide two specific examples from the scenario that demonstrate the consequences of overlooking contextual factors. 2

(c) Highlight at least two practices that may have contributed to the operational issues and discuss how alternative practices could have prevented these problems. 3

(d) Reflecting on the scenario, discuss two key lessons that software maintenance practitioners can learn from the challenges faced by ABC Tech Inc. in the deployment of the inventory management system. 2

2\. (a) Identify and explain two principles of software maintenance that ABC Tech Inc. may have violated during the development and deployment of the inventory management system. 4

How do these violations contribute to the issues encountered after the system's full-scale implementation?

(b) Analyze the importance of code safety and reliability in the context of the inventory management system scenario. 3

(c) In the context of the scenario where ABC Tech Inc.'s inventory management system introduced new issues after full-scale implementation, what problems 3 might maintainers face when rewriting or reengineering the inventory management system? What are the causes of those problems?

3. Suppose, you work at an e-commerce company named Daraz. One day, a client reports a critical bug in the payment processing system, causing transactions to fail. This issue needs immediate attention to prevent financial losses. Meanwhile, another client reports a minor issue with the website's search functionality, which does not affect sales but impacts user experience.

Now, consider the scenario to answer the following questions:

(a) How would you manage and resolve this customer problem quickly to minimize service interruption? 4

(b) How you should handle this urgent payment problem differently than a less urgent problem, like a small search issue on the website? 3

(c) Give an example of a small, non-urgent problem that might happen in an online store like Daraz and explain how you should decide when to fix it and what steps you should take. 3

4. Your company, Samsung R&D, is undertaking a major project to update a 20-year-old inventory management system to a modern architecture. This system is critical for tracking and managing inventory across multiple warehouses and must remain operational during the update process. You need to plan this extensive evolution carefully to ensure continuity of service and data integrity.

Now, consider the scenario to answer the following questions:

(a) Describe the challenges you may face in updating this system to a modern architecture while keeping it operational. 4

(b) What techniques would you use to manage the evolution of this system? You must focus on minimizing downtime and ensuring data integrity. 3

(c) How would you use architectural patterns to isolate changes and add new functionalities gradually? Provide examples of the patterns and their benefits that you would use. 3

5. You are a DevOps engineer at InnovateSoft, a software development company specializing in enterprise solutions. Your team has been tasked with maintaining and enhancing a critical inventory management system for a large retail chain, MegaStores Inc. This system handles millions of transactions daily, and any downtime or issues can significantly impact the business.

Recently, the system has faced several challenges:

- There have been issues with reusing existing components due to tight coupling and complex dependencies.
- The system requires restructuring to improve maintainability and performance.
- Monitoring and logging mechanisms are inadequate, leading to difficulties in collecting, interpreting, and visualizing data.

Deployment processes have been problematic, causing delays and errors during releases.

(a) Suggest three strategies that your team can implement to improve the reuse of existing components in the inventory management system. 3

(b) Explain how centralized logging can help in collecting data more effectively and discuss potential problems with this approach. 3

(c) Propose a detailed plan for restructuring the inventory management system to address the tight coupling issue. Include key steps and best practices. 4

6 (a) Define software rejuvenation 2

(b) Explain plan recognition. 2

(c) Explain how "bad smells" in code interfere with maintainability. 3

(d) "The maintainability of the system should not degrade. Otherwise, future changes
Will be more expensive to carry out." Evolute the statement. 3

OR

(a) Explain the concept of specification recovery and its importance in reverse engineering. 3

(b) Illustrate with an example how design recovery can be used to understand legacy systems. 3

(c) Discuss the steps involved in the reengineering process and the benefits it offers to software maintenance and evolution. 4

7 (a) Define what is meant by "target for reuse" in software engineering. Explain with proper example. 3

(b) Explain the benefits of software reusability to both the development team and the end-users. 3

(c) Outline the steps involved in domain analysis and the outcomes of each step. 4

OR

(a) What is software aging? What are the common causes of software aging? How can those causes be eliminated? Discuss briefly. 4

(b) What is software entropy? Explain its relationship with software aging. 3

(c) Explain the term software cloning and discuss its significance concerning software evolution. 3

Good Luck

Institute of Information Technology (IIT)

Noakhali Science and Technology University

Bachelor of Science in Software Engineering
Final Examination-2022 (Year-4, Term-2)

Course Title: Software Project Management
Course Code: SE4205

Total Time: 3 Hours
Full Marks: 70

Answer any Seven (07) questions from the following: 7 X 10 = 70

1. a. A Software house has developed a customized order processing system for a client. You are an employee of the software house that has been asked to organize a training course for the end-users of the system. At present, a user handbook has been produced, but no specific training material. A plan is now needed for the project which will set up the delivery of the training courses. The project can be assumed to have been completed when the first training course starts. Among the things that will need to be considered are the following: 8

- training materials will need to be designed and created
- a timetable will need to be drafted and agreed
- date(s) for the course will need to be arranged
- the people attending the course will need to be identified and notified
- Rooms and computer facilities for the course will need to be provided.

a) Identify the main stakeholders for the project
b) Draw up objectives for this project
c) For the objectives, identify the measures of effectiveness.
d) For each objectives, write down sun-objectives or goals and the stakeholders who will be responsible for their achievement.

b. Describe briefly about the main responsibility of a software project manager. 2

2. a. 6

Year	Project1	Project2	Project3	Project4
0	-100.000	-1.000.000	-100.000	-120.000
1	10.000	200.000	30.000	30.000
2	10.000	200.000	30.000	30.000
3	10.000	200.000	30.000	30.000
4	20.000	200.000	30.000	30.000
5	100.000	300.000	30.000	75.000
Net Profit	50.000	100.000	50.000	75.000

Brightmouth College is considering the replacement of the above existing payroll service, operated by a third party, with a tailored, off-the-shelf computer-based system. List some of the costs it might consider under the headings of:

- Development costs
- Setup costs
- Operational costs

List some of the benefits under the headings:

- Quantified and valued benefits
- Quantified but not valued
- Identified but not easily valued

employee [illegible], efficiency

For each cost or benefit, explain how, in principle, it might be measured in monetary terms.

b. Consider the four project cash flows given the above table and calculate the payback period for each of them. 2

c. Calculate the Return on investment (ROI) for above projects (1-4) and decide which, on the basis of this criterion, is the most worthwhile. 2

3. a. An employee of a training organization has the task of creating case study exercises and solutions for a training course which teaches a new systems analysis and design method. The person's work plan has a three-week task 'learn new method'. A colleague suggests that this is unsatisfactory as a task as there are no concrete deliverables or products from the activity. What can be done about this? 5

b. In order to carry out usability tests for a new word processing package, the software has to be written and debugged. User instructions have to be available describing how the package is to be used. These have to be scrutinized in order to plan and design the tests. Subjects who will use the package in the tests will need to be selected. As part of this selection process, they will have to complete a questionnaire giving details of their past experience of, and training in, typing and using word processing packages. The subjects will carry out the required tasks using the word processing package. The tasks will be timed and any problems the subjects encounter with the package will be noted. After the test, the subjects will complete another questionnaire about what they felt about the package. All the data from the tests will be analyzed and a report containing recommendations for changes to the package will be drawn up. Draw up a Product Breakdown Structure, a Product Flow Diagram and a preliminary activity network for the above. 5

Suppose, an E-commerce shop wants to introduce new **gift vouchers** for their customers. With this card customers will get discounts and get some offer on different products. These vouchers should be valid for one year from purchase and it can be refilled. As, the online shopping market is getting competitive day by day they have taken this strategy though it may narrow down their profit margin. However, they have set some milestones regarding this to enrich their brand value and rule the market.

a. Analyze the above scenario using SWOT approach. 4

b. Identify the potential risks for the E-Commerce shop? Show the risks calculating the risk score based on likelihood and consequence. 3

c. For this new concept of the E-commerce shop they need to emphasize their employees. Therefore, the project manager is asked to motivate the team for successful completion of the above project. 3

Now, think yourself as a Project Manager. What techniques you will do to motivate the team? Justify your answer.

5. a. In order to carry out usability tests for a new word processing package, the software has to be written and debugged. User instructions have to be available describing how the package is to be used. These have to be scrutinized in order to plan and design the tests. Subjects who will use the package in the tests will need to be selected. As part of this selection process, they will have to complete a questionnaire giving details of their past experience of, and training in, typing and using word processing packages. The subjects will carry out the required tasks using 5

the word processing package. The tasks will be timed and any problems the subjects encounter with the package will be noted. After the test, the subjects will complete another questionnaire about what they felt about the package. All the data from the tests will be analyzed and a report containing recommendations for changes to the package will be drawn up. Draw up a Product Breakdown Structure, a Product Flow Diagram and a preliminary activity network for the above.

b. Describe about the scope and role of change management in software project management. 3

c. Why is stakeholder management important in software project management? 2

6. Considering the Data below: 10

Project	Inputs	Outputs	Entity accesses	System users	Programming Language	Developer days
1	210	420	40	10	X	30
2	469	1406	125	20	X	85
3	513	1283	76	18	Y	108
4	660	2310	88	200	Y	161
5	183	367	35	10	Z	22
6	244	975	65	25	Z	42
7	1600	3200	237	25	Y	308
8	582	874	111	5	Z	62
X	180	350	40	20	Y	
Y	484	1190	69	35	Y	

a. What items are productivity drivers?

b. What are the productivity rates for programming languages x, y and z?

c. What would be the estimated effort for projects X and Y using a Mark II function point count?

d. What would be the estimated effort for X and Y using an approximate analogy approach?

e. What would have been the best estimating method if the actual effort for X turns out to be 30 days and for Y turns out to be 120 days? Can you suggest why the results are as they are and how they might be improved?

7. ***Scenario (Question 7-8)***

Consider the following table for a particular Software Development project.

Task No.	Task Type	Task Name	Predecessors Tasks (Dependencies)	Time (Days)	Human Resource
A	*Requirement*	Requirement Elicitation	-	5	3
B	*Analyst*	System Analysis	-	7	2
C	*Design*	UI Design	A	6	3
D	*DB*	Database Design	A	4	2
E	*QA*	Test Case & Test Plan	B	3	5
F	*Development*	User Access Control	E	2	3
G	*Development*	Additional Functionalities	C	4	4

H	*Development*	Admin Panel	D, F	5	5
I	*Development*	Reporting Module	E	4	5
J	*Testing*	White Box Testing	G	3	4
K	*Deployment*	Black Box Testing	J, H	4	3
L	*Training*	User Training	I	6	4

a. Draw the PERT chart for the above tasks considering the predecessors and required time. 5

b. Identify the tasks that are on the Critical Path of the PERT chart? 3

c. Draw the Gantt Chart mentioning the dependencies of the tasks. 2

8. a. A software package is to be designed and built to assist in software cost estimation. It will input certain parameters and produce initial cost estimates to be used at bidding time. 4

I. It has been suggested that a software prototype would be of value in these circumstances. Explain why this might be.

II. Discuss how such prototyping could be controlled to ensure that it is conducted in an orderly and effective way and within a specified time span.

b. What stage of a system development project (for example, feasibility study, requirements analysis etc.) would a prototype be useful as a means of reducing the following uncertainties? 6

i. A computer system is to support sales office staff taking phone calls from members of the public enquiring about motor insurance and giving quotations over the phone.

ii. The insurance company is considering implementing the telephone sales system using the system development features supplied by Microsoft Access' They are not sure, at the moment, that it can provide the kind of interface that would be needed and are also concerned about the possible response times of a system developed using Microsoft Access.

9. Below are details of a project. All times are in days.

Activity	Depends on	Most likely	Plus safety
A		10	14
B	A	5	7
C	B	15	21
D	A	3	5
E	A	8	12
F	E	20	22
G	D	6	8
H	C,F,G	10	14

a. Using (i) the most likely and then (ii) the safety estimates: 5

- Calculate the earliest and latest start and end days and float for each activity.
- Identify the critical path.

b. Draw up an activity diagram applying critical chain principles for this project; 5

- Locate the places where buffers will need to be located.
- Assess the size of the buffers.
- Start all activities as late as possible

End

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Noakhali Science and Technology University
Bachelor of Science in Software Engineering
Final Examination - 2022 (Year-4, Term-2)

Course Title: Applied Data Science **Total Time:** 3 Hours
Course Code: CSE 4213 **Full Marks:** 70

Answer any Seven (07) questions from the following: **7 X 10 = 70**

1. Consider the following dataset which will be used to predict if students pass in Applied Data Science course (Yes or No), based on their previous GPA (High, Medium, or Low) and whether or not they studied.

GPA	Studied	Passed		GPA	Studied	Passed
Low	False	No		Medium	True	Yes
Low	True	Yes		High	False	Yes
Medium	False	No		High	True	Yes

a. Calculate the entropy of the output variable, *H(Passed)*. 3

b. Compute the information gain of initially choosing the variable GPA. 3

c. Construct the full decision tree that would be learned for this dataset. Show necessary calculations. 4

2. A professor wants to know if students are getting enough sleep. Each day, the professor observes whether the students have red eyes in the class. The professor has the following domain knowledge.

- The prior probability of getting enough sleep, with no observations, is 0.7.
- The probability of getting enough sleep on night t is 0.8 given that the student got enough sleep the previous night, and 0.3 if not.

The probability of having red eyes is 0.2 if the student got enough sleep, and 0.7 if not.

a. Formulate this information as a dynamic Bayesian network that the professor could use to filter or predict from a sequence of observations. 3

b. Reformulate it as a hidden Markov model for this observation variable. Give the complete probability tables for the model. 3

c. Calculate the probability that a student has red eyes in *Day 3* given that he has not slept well in *Day 2*. 4

3. Consider a binary classification task with the following training data.

Class A: (1.0, 1.0), (2.0, 1.0), (2.5, 2.0), (1.5, 4.0)
Class B: (2.0, 2.5), (3.0, 3.0), (0.0, 0.0), (1.0, 3.5)

a. Apply 3-Nearest Neighbors algorithm (using Euclidean distance) to identify the label of a test point (2.0, 2.0) before and after inserting another training point (2.5, 2.5) that belongs to Class B. 4

b. For a binary classification task, what will be the training error for a 1-Nearest Neighbor? Briefly explain your answer. 3

c. Discuss why *k*-Nearest Neighbors is called a lazy learner. 3

4\. a. What is the general principle of an ensemble method? What is bagging and boosting in ensemble method? 3

b. "Accuracy is a good performance metric for imbalanced dataset"- Do you agree with this statement? Justify your answer with an example. 3

c. Mr. Gibson has built a random forest model with 10000 trees. He has got delighted after getting training error as 0.00. However, the validation error is 34.23. What is going on? What might be probable mistakes of Mr. Gibson has made? 4

5\. a. What do you understand by under fitting and over fitting? 2

b. For a binary classification problem, a model has been used to predict the class label ten samples. The actual class labels and predicted labels are given in the following table. 4

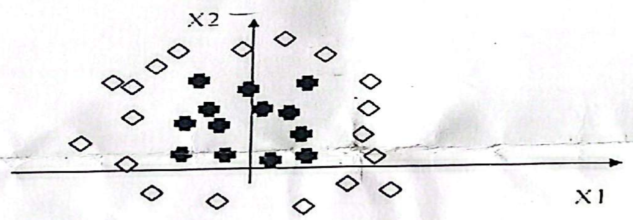
Actual Label	1	1	1	0	0	0	1	1	0	0
Predicted Label	1	0	1	0	0	1	0	1	1	1

FN FP

Create a confusion matrix for the given data. Also calculate accuracy and F1 score.

c. Draw the receiver operating characteristics (ROC) curve for the aforementioned situation. Also compute the area under ROC curve. 4

6\. a. Suppose that Mr. Samson wants to build a neural network that classifies two dimensional data (*i.e.*, $X = [x1, x2]$) into two classes: diamonds and crosses. We have a set of training data that is plotted as follows. 5



TPR

Draw a network that can solve this classification problem. Justify your choice of the number of nodes and the architecture. Draw the decision boundary that your network can find on the diagram.

b. Reconsider the figure given in part (a). Discuss why hard margin linear Support Vector Machine (SVM) fails to separate the instances of two classes. Also explain how kernel tricks assist to find separable hyperplane in this case. 5

7\. The heights of 8 randomly chosen sons versus that of their fathers (in inches) are as follows.

Father (*X*)	60	62	64	65	66	67	68	70
Son (*Y*)	63.6	65.2	66	65.5	66.9	67.1	67.4	68.3

a. Estimate the regression coefficients based on the given data. 3

b. Find the coefficient of determination. And comment on how the height of a son relates to that of his father. 4

c. Discuss a situation when you choose regression over classification. 3

8. a. Consider the dataset: {0, 5, 6, 20, 25, 38, 43, 44}. 6

Mr. Rick wants to identify the two top-level clusters from his dataset. What results will he get using single link, complete link, and average link clustering methods? If single link and complete link give the same two clusters, does it indicate that average link will output the same two clusters? Explain with this example.

b. The table below presents a distance matrix for 6 objects. 4

	A	B	C	D	E	F
A	0					
B	0.12	0				
C	0.51	0.25	0			
D	0.84	0.16	0.14	0		
E	0.28	0.77	0.70	0.45	0	
F	0.34	0.61	0.93	0.20	0.67	0

i. Show the final result of hierarchical clustering with single link by drawing a dendrogram.

ii. Show the final result of hierarchical clustering with complete link by drawing a dendrogram.

9. a. Describe k-fold cross validation method in brief. 3

b. Define core point, boundary point, and noise point for density-based clustering. 3

c. Dr. John Kenedy has applied hierarchical clustering algorithm and got two clusters, $C_1 = \{(1, 1), (2, 2), (3, 1)\}$ and $C_2 = \{(5, 3), (6, 4)\}$. Now he wants to approximate the distance between these two clusters (using Euclidean distance). 4

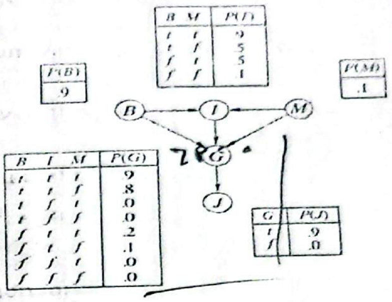
Calculate the distance between C_1 and C_2 applying single link, complete link, and average link distance measure.

Good Luck!

Institute of Information Technology
Noakhali Science and Technology University
Bachelor of Science in Software Engineering
4th Year 2nd Semester, Session: 2018-19 (Quiz-1)

Course Title:	Applied Data Science	**Total Time:**	1 Hour
Course Code:	CSE 4213	**Full Marks:**	30

1. Consider the simple Bayesian network with boolean variables where B indicates Broke Election Law, I indicates Indicted, M indicates Politically Motivated Prosecutor, G indicates Found Guilty, and J indicates Jailed.



a. Which of the following equations can be asserted by the network *structure*? Give proper explanation for each of these. 6

(i) P(B, I, M) = P(B)P(I)P(M).
(ii) P(J | G) = P(J | G, I).
(iii) P(M | G, B, I) = P(M | G, B, I, J)

b. Calculate the probability that someone goes to jail given that he broke the law, has been indicted, and faced a politically motivated prosecutor. 4

2. A professor wants to know if students are getting enough sleep. Each day, the professor observes whether the students have red eyes in the class. The professor has the following domain knowledge.

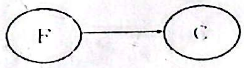
- The prior probability of getting enough sleep, with no observations, is 0.7.
- The probability of getting enough sleep on night *t* is 0.8 given that the student got enough sleep the previous night, and 0.3 if not.
- The probability of having red eyes is 0.2 if the student got enough sleep, and 0.7 if not.

a. Formulate this information as a dynamic Bayesian network that the professor could use to filter or predict from a sequence of observations. 3

b. Reformulate it as a hidden Markov model for this observation variable. Give the complete probability tables for the model. 4

c. Calculate the probability that a student has red eyes in *Day 5* given that he has not slept well in *Day 4*. 3

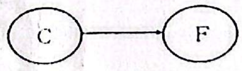
3. Consider the following Bayesian network where F = having the flu and C = coughing.

P(F) = 0.1


P(C|F) = 0.8
P(C|¬F) = 0.3

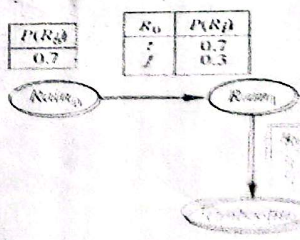
a. Write down the joint probability table specified by the Bayesian network. 2

b. Determine the probabilities for the following Bayesian network so that it specifies the same joint probabilities as the given one. 3



By looking at the conditional probability table, do you think C and F are independent? Provide a rationale for your response.

c. Consider the dynamic Bayesian network for the Umbrella World scenario given in the figure. 5



Suppose you observe an unending sequence of days on which the umbrella appears. Show that, as the days go by, the probability of rain on the current day increases monotonically toward a fixed point. Calculate this fixed point of appearing an umbrella.